

1 Summary of Features

The TC33xEXT product family has the following features:

- High Performance Microcontroller with two CPU cores
- Two 32-bit super-scalar TriCore CPUs (TC1.6.2P), each having the following features:
 - Superior real-time performance
 - Strong bit handling
 - Fully integrated DSP capabilities
 - Multiply-accumulate unit able to sustain 2 MAC operations per cycle
 - Fully pipelined Floating point unit (FPU)
 - up to 300 MHz operation at full temperature range
 - up to 192 Kbyte Data Scratch-Pad RAM (DSPR)
 - up to 64 Kbyte Instruction Scratch-Pad RAM (PSPR)
 - up to 64 Kbyte Data RAM (DLMU)
 - 32 Kbyte Instruction Cache (ICACHE)
 - 16 Kbyte Data Cache (DCACHE)
- Lockstepped shadow core for one TC1.6.2P
- Multiple on-chip memories
 - All embedded NVM and SRAM are ECC protected
 - up to 2 Mbyte Program Flash Memory (PFLASH)
 - up to 128 Kbyte Data Flash Memory (DFLASH 0) usable for EEPROM emulation
 - BootROM (BROM)
- 64-Channel DMA Controller with safe data transfer
- Sophisticated interrupt system (ECC protected)
- High performance on-chip bus structure
 - 64-bit Cross Bar Interconnect (SRI) giving fast parallel access between bus masters, CPUs and memories
 - 32-bit System Peripheral Bus (SPB) for on-chip peripheral and functional units
 - SRI to SPB bus bridges (SFI Bridge)
- Optional Hardware Security Module (HSM) on some variants
- Safety Management Unit (SMU) handling safety monitor alarms
- Memory Test Unit with ECC, Memory Initialization and MBIST functions (MTU)
- Versatile On-chip Peripheral Units
 - 6 Asynchronous/Synchronous Serial Channels (ASCLIN) with hardware LIN support (V1.3, V2.0, V2.1 and J2602) up to 50 MBaud
 - 4 Queued SPI Interface Channels (QSPI) with master and slave capability up to 50 Mbit/s
 - 1 MCMCAN Modules with 4 CAN nodes for high efficiency data handling via FIFO buffering
 - 6 Single Edge Nibble Transmission (SENT) channels for connection to sensors
 - One Capture / Compare 6 module (Two kernels CCU60 and CCU61)
 - One General Purpose 12 Timer Unit (GPT120)
 - 1 IEEE802.3 Ethernet MAC with RMII and MII interfaces (ETH)
- Versatile Successive Approximation ADC (VADC)
 - Cluster of 6 independent ADC kernels

Summary of Features

- Input voltage range from 0 V to 5.5V (ADC supply)
- Digital programmable I/O ports
- On-chip debug support for OCDS Level 1 (CPUs, DMA, On Chip Buses)
- Multi-core debugging, real time tracing, and calibration
- Four/five wire JTAG (IEEE 1149.1) or DAP (Device Access Port) interface
- Power Management System and on-chip regulators
- Clock Generation Unit with System PLL and Peripheral PLL
- Embedded Voltage Regulator
- Qualified for automotive application according to AEC-Q100 (only applicable after delivery release of the corresponding sales codes)
- ISO 26262 Safety Element out of Context for safety requirements up to ASIL D (only applicable for sales codes listed within a released Safety Package Release Note from IFX)

Summary of Features

Ordering Information

The ordering code for Infineon microcontrollers provides an exact reference to the required product. This ordering code identifies:

- The derivative itself, i.e. its function set, the temperature range, and the supply voltage
- The package and the type of delivery.

Table 1-1 Platform Feature Overview

Feature		TC33xEXT
CPUs	Type	TC1.6.2
	Cores / Checker Cores	2 / 1
	Max. Freq.	300 MHz
Cache per CPU	Program	32 KB
	Data	16 KB
SRAM per CPU	PSPR	32 KB for CPU 0/ 64 KB for CPU 1
	DSPR	192 KB for CPU0/ 96 KB for CPU1
	DLMU	8 KB for CPU0/ 64 KB for CPU1
Extension Memory	TCM	1 MB
	XCM	-
	XTM	16 KB
Program Flash	Size	2 MB
	Banks	1 x 2 MB
Data Flash	Size (single-ended)	128 KB (DF0) + 128 KB (DF1)
DMA	Channels	64
CONVCTRL	Modules	1
EVADC	Primary Groups/Channels	6 / 40
	Secondary Groups/Channels	0 / 0
	Fast Compare Channels	-
EDSADC	Channels	-
GTM	Clusters	0
	TIM (8 ch)	0
	TOM (16 ch)	0
	ATOM (8 ch)	0
	MCS (8 ch)	0
	CMU / ICM	0 / 0
	PSM	0
	TBU channels	0
	SPE	0
	CMP / MON	0 / 0
	BRC / DPLL	0 / 0
	CDTM modules	0
	DTM modules	0

Summary of Features

Table 1-1 Platform Feature Overview (cont'd)

Feature		TC33xEXT
Timer	GPT12	1
	CCU6	1 module with 2 kernels
STM	Modules	2
FlexRay	Modules	0
	Channels	0
CAN	Modules	1
	Nodes	1 x 4
	of which support TT-CAN	0
QSPI	Modules	4
	HSIC Channels	2
ASCLIN	Modules	6
I2C	Interfaces	0
SENT	Channels	6
PSI5	Modules	0
PSI5-S	Modules	0
HSSL	Channels	0
MSC	Channels	0
EBU	External Bus	0
SDMMC	eMMC/SD Interface	1
Ethernet (10/100Mbit/1Gbit)	Modules	1
FCE	Modules	1
Safety Support	SMU	yes
	IOM	no
SPU	Modules	1
RIF	Modules	1
HSPDM	Modules	1
Security	HSM+	1
Debug	OCDS	yes
	MCDS	no
	miniMCDS	no
	miniMCDS TRAM	no
	MCDS light	yes
	AGBT ¹⁾	yes
Low Power Features	Standby RAM	1 - DLMU0
	SCR	yes
Packages	Type	LFBGA-292 / LFBGA-180
I/O	Type	5 V CMOS / 3.3 V CMOS / LVDS
T _{ambient}	Range	-40 ... +125°C

Summary of Features

1) The Aurora Gigabit Trace Module (AGBT) is a trace interface intended for development use only (not to be used in series production). It is only available on the specific emulation devices with feature package E, T, and on ADAS devices with feature package A, H of the TC39x, TC37xEXT, TC35x, TC33xEXT. AGBT I/O functions are only available for packages with 292 or more pins. For details on AGBT parameters see the "TC3xx Emulation Devices" Data Sheet.

2 TC33xEXT Pin Definition and Functions:

The following figures are showing the TC33xEXT Logic Symbols for the package variants:

- LFBGA-292 for feature package DA and DZ ([Figure 2-1](#))
- LFBGA-180 for feature package DA ([Figure 2-2](#))

TC33xEXT Pin Definition and Functions:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			
A	NC1	VEXT	NC	NC	NC	NC	NC	NC	NC	P11,11	P11,9	P11,2	P14,8	P14,5	P14,1	P15,6	P15,4	P15,1	VDDP3	VSS	A		
B	NC	VSS	VEXT	NC	NC	NC	NC	NC	NC	P11,12	P11,10	P11,3	P14,6	P14,3	P14,4	P14,0	P15,3	VDDP3	VSS	P15,0	B		
C	P50_1	P50_0																	P15,2	P20,14	C		
D	P50_3	P50_2	VSS	VFLEX	P11,15	P11,14	P11,5	P11,6	P11,4	P14,10	P14,9	P14,7	P15,8	P15,7	VDD	VSS			P20,12	P20,13	D		
E	P50_5	P50_4	P10,0	VSS	P11,13	P11,8	P11,7	P11,1	P11,0	P12,1	P12,0	P14,2	P15,5	VDD	VSS	P20,9			P20,10	P20,11	E		
F	P50_7	P50_6	P10,1	P10,2											ESR0	P20,6			P20,7	P20,8	F		
G	P50_9	P50_8	P10,3	P10,4											ESR1	FORST			P20,1	P20,3	G		
H	P50_11	P50_10	P10,5	P10,6	VDD		VSS	VSS	VSS	VSS		VDD			P21,7 / TDO	P21,6 / TDI			P20,2	P20,0	H		
J	P02,0	P02,1	P10,8	P10,7	VSS	VSS		VSS	VSS		VSS	VSS			TCK	P21,1			P21,3	P21,5	J		
K	P02,2	P02,3	P02,4	P02,5	AGBTC LKP (VSS)	VSS	VSS	VSS	VSS	VSS	VSS	DAPE0			TMS	P21,0			P21,2	P21,4	K		
L	P02,6	P00,0	P02,7	P02,8	AGBTC LKN (VSS)	VSS	VSS	VSS	VSS	VSS	VSS	AGBTE RR (VSS)			NC	NC			TRST	VSS	L		
M	P00,2	P00,1	P00,3	P00,4	VSS	VSS		VSS	VSS		VSS	VSS			NC	NC			XTAL2	XTAL1	M		
N	P00,8	P00,7	P00,6	P00,5	VDD		VSS	VSS	VSS	VSS		VDD			NC	NC			VDD	VEXT	N		
P	NC	P00,9	NC	NC											P22,4	P22,5			P22,1	P22,0	P		
R	NC	P00,12	NC	AN23											NC	NC			P22,3	P22,2	R		
T	NC	NC	NC	AN22	AN15	AN12	AN6	AN4	AN0	VEVRS B	P34,2	NC	NC	NC	VSS	P23,5			NC	NC	T		
U	NC	NC	NC1	AN17	AN14	AN9	AN7	AN3	AN1	P34,1	P34,3	NC	NC	NC	NC	VSS			P23,1	NC	U		
V	NC	NC																	VEXT	NC	V		
W	NC	NC	AN19	AN18	AN16	AN13	AN11	AN8	AN2	P33,0	P33,2	P33,4	P33,6	P33,8	P33,10	P33,12	P32,1 / VGATE 1P	P32,4	VSS	VEXT	W		
Y	NC1	AN21	AN20	VSSM	VDDM	VAREF 1	VAGND 1	AN10	AN5	P33,1	P33,3	P33,5	P33,7	P33,9	P33,11	P33,13	P32,0 / VGATE 1N	NC	NC	VSS	Y		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			

Figure 2-1 TC33xEXT Logic Symbol for the package variant LFBGA-292 for feature package DA and DZ

TC33xEXT Pin Definition and Functions:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NC	NC	P11.10	P11.9	P11.6	P11.2	P11.0	P12.1	P14.0	P15.3	P15.6	P15.1	P15.0	NC	A
B	NC	VSS	P11.8	P11.7	P11.5	P11.4	P11.1	P11.14	P14.5	P14.1	P15.4	P15.2	VSS	P20.14	B
C	P50_0	P50_1	VSS	P11.12	P11.3	P11.15	P11.13	P12.0	P14.6	P14.4	P14.3	VSS	P20.13	P20.10	C
D	P50_2	P50_3	NC	VSS	VFLEX	P11.11	NC	NC	P14.2	P15.5	VSS	P20.9	P20.12	P20.11	D
E	P50_4	P50_5	P10.1	P10.6	VSS					VSS	ESR1	P20.7	PORST	ESR0	E
F	P50_6	P50_7	P10.2	P10.5		VEXT	VDDP3	VEXT	VDD		P20.6	P20.8	P21.6 / TDI	P21.7 / TDO	F
G	P50_8	P50_9	P10.8	P10.3		VEXT	VSS	VSS	VDD		TMS	P20.3	P20.2	TCK	G
H	P50_10	P50_11	P10.7	P02.1		VDD	VSS	VSS	VEVRS B		NC	P21.0	P20.0	P21.5	H
J	P02.4	P02.5	P02.2	AN17		VSS	VDD	VDD	VEXT		NC	NC	P21.4	P21.3	J
K	P02.0	P02.7	AN20	AN16	VSS					VSS	NC	TRST	P21.2	VSS	K
L	P02.8	P02.6	NC	AN9	AN4	AN2	NC	P33.4	P33.8	P33.7	VSS	NC	XTAL2	XTAL1	L
M	P02.3	P00.0	AN21	AN12	AN5	AN3	NC	P33.1	P33.0	P33.12	P33.13	VSS	VDD	VEXT	M
N	AN15	AN14	AN13	AN11	AN8	AN0	NC	P33.3	P33.5	P33.11	P33.10	P32.1 / VGATE 1P	VSS	P23.1	N
P	NC	VSSM / VAGND 1	VDDM	VAREF 1	AN10	AN1	NC	P33.2	P33.6	P33.9	VEXT	P32.0 / VGATE 1N	P32.4	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

Figure 2-2 TC33xEXT Logic Symbol for the package variant LFBGA-180 for feature package DA